

SHREYAS SANGHVI

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EDUCATION

Carnegie Mellon University , Pittsburgh, PA	Expected. Dec 2025
Master of Science in Biomedical Engineering - Research Track	GPA: 4.0/4.0
Award: Biomedical Engineering Department Head's Fellowship, Scholarly Project Funding – Spring 2025	
NIIT University , Neemrana, India	
Bachelor of Technology in Electronics and Communication with <i>Distinction</i> .	CGPA: 8.81/10
Award: Ram Rajinder Malhotra Medal for best all-around graduating student	

SKILLS

Programming and databases: C/C++, Embedded C, Java, Python, SQL, Neo4J
Hardware: Advance: PIC16F family, STM32F1/F4 families; *Beginner:* TI C2000ware, TI F2806x family
Software: Eagle CAD, OrCAD, Altium, MATLAB, LTSpice, STM32 Cube, Kiel uVision, Solidworks, Labview, NI DAQmx, Linux, Embedded Linux
Prototyping: 3D printing (FDM / SLA), Laser cutting, Laser welding, Wood working

EXPERIENCE

R&D Intern <i>IotaMotion, Pittsburgh, PA</i>	May 2025 - Aug 2025
- Designed and assembled electro-mechanical system for real-time surgical force feedback, adhering to IEC 60601-1 - Reduced fault recovery time by 20% through optimizing hardware-software feedback control, ensuring ISO 13485 and IEC 62304 compliance - Automated motor parameter validation with custom test rigs and embedded scripts, boosting testing efficiency by 15%, aligned with IEC 62304 and ISO 13485 compliance	
Founder <i>Phystech Labs Private Limited, Mumbai, India</i>	Nov 2020 - Aug 2024
- Raised INR 3 million to develop a feedback system with applications in diabetic foot ulcer management - Coordinated design and firmware development of graphene-based smart sock for real-time foot pressure monitoring, improving mobility in 150+ patients, compliant with IEC 62304 and ISO 13485 standards - Engineered real-time pressure feedback algorithms to enable precise offloading guidance, accelerating patient recovery by 25% and optimizing clinical outcomes	
Contract Engineer <i>Jiva Sciences Private Limited, Bangalore, India</i>	Aug 2022 - Apr 2023
- Implemented a microfluidics control system using TI-F28069M microcontrollers and OrCAD design software - Reduced CNC micro-milling control box footprint by 10% through layout optimization	

PROJECTS

HeartPrinter: A Parallel Wire Robot for Cardiac Intervention Github	Aug 2024 - Present
<i>Carnegie Mellon University, Pittsburgh, PA</i> - Enhancing control loop using NI DAQmx to achieve faster and more accurate sensor feedback, improving system localization and navigation during cardiac procedures - Integrating real-time navigation capabilities enabling autonomous tracking and compensation for heart motion during beating-heart procedures - Redesigning the platform base to increase mechanical stability and precision under dynamic operating loads	
FALCON: FoveA LoCalizatiON in En-face OCT Imaging via Explainable B-scan Classification and Transformer-Based Segmentation Github	Jan 2025 - Present
<i>Carnegie Mellon University, Pittsburgh, PA</i> - Proposed a dual pipeline machine learning algorithm as a team of four, with the target of automating fovea localization in En-Face Optical Coherence Tomography (OCT) images - Leveraging VGG16 based classification on OCT B-Scans and Pretrained MAE autoencoder with U-Net++ decoder to achieve 98.72% accuracy on validation data set of B-Scan OCT images	
Dr. Phix-it	Oct 2024
<i>BioHacks 2024, Nucleate Pittsburgh, PA</i> - Led a team of four engineers to design a mobile interface in under 24 hours that demonstrates real-time data augmentation using haptics and audio-visual cues for cochlear implant surgeries - Secured First place at BioHacks 2024 and received a prize of US\$1500	